KAIZEN MATHS

Lesson starter

Percentage Applications: Finance and Contexts

Advanced Finance

Name _____

S1

£10,949.00 is invested at 6.84% compound interest for 6 years. Find the final amount and interest earned.

S2

A mortgage of £331,259.00 at 6.29% annually for 15 years. Calculate the monthly payment and total interest paid.

S3

An investment of £12,938.00 grew to £20,966.92 over 5 years. Calculate the total return percentage and annual return rate.

S4

£20,732.00 is invested at 4.43% compound interest for 3 years. Find the final amount and interest earned.

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NUM-07B4-L3 · S

Show your working

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Exit ticket

Percentage Applications: Finance and Contexts

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E1

A mortgage of £283,275.00 at 6.4% annually for 20 years. Calculate the monthly payment and total interest paid.

E2

An investment of £41,382.00 grew to £94,384.31 over 6 years. Calculate the total return percentage and annual return rate.

E3

£5,801.00 is invested at 4.92% compound interest for 10 years. Find the final amount and interest earned.

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NUM-07B4-L3 · E

Show your working

Teacher answers & working

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S1

£10,949.00 is invested at 6.84% compound interest for 6 years. Find the final amount and interest earned.

Answer:

Final amount: £16,284.62, interest: £5,335.62

Working:

Use $A = P(1 + r)^t$.

$P = £10{,}949.00$, $r = 0.0684$, $t = 6$

Growth multiplier = 1 + 0.0684 = 1.0684

$A = £10{,}949.00 \times (1.0684)^6$

$A = £10{,}949.00 \times 1.4873 = £16{,}284.62$

Interest = £16,284.62 - £10,949.00 = £5,335.62

S2

A mortgage of £331,259.00 at 6.29% annually for 15 years. Calculate the monthly payment and total interest paid.

Answer:

Monthly payment: £2,847.52, total interest: £181,294.60

Working:

Monthly rate = 6.29% ÷ 12 = 0.5242% = 0.005242

Number of payments = 15 × 12 = 180

$(1 + r)^n = (1 + 0.005242)^{180} = 2.5626$

$M = P \times [r(1+r)^n] \div [(1+r)^n - 1]$

$M = £331{,}259.00 \times [0.005242 \times 2.5626] \div [2.5626 - 1]$

Monthly payment = £2,847.52

Total paid = £2,847.52 × 180 = £512,553.60

Total interest = £512,553.60 - £331,259.00 = £181,294.60

S3

An investment of £12,938.00 grew to £20,966.92 over 5 years. Calculate the total return percentage and annual return rate.

Answer:

Total return: 62.06%, annual return: 10.14%

Working:

Total return = £20,966.92 - £12,938.00 = £8,028.92

Total return % = (£8,028.92 ÷ £12,938.00) × 100 = 62.06%

Annual return = $[(£20{,}966.92 \div £12{,}938.00)^{1 \div 5} - 1] \times 100$

Annual return = 10.14%

Teacher answers & working

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S4

£20,732.00 is invested at 4.43% compound interest for 3 years. Find the final amount and interest earned.

Answer:

Final amount: £23,611.14, interest: £2,879.14

Working:

Use $A = P(1 + r)^t$.

$P = £20,732.00$, $r = 0.0443$, $t = 3$

Growth multiplier = 1 + 0.0443 = 1.0443

$A = £20,732.00 \times (1.0443)^3$

$A = £20,732.00 \times 1.1389 = £23,611.14$

Interest = £23,611.14 - £20,732.00 = £2,879.14

E1

A mortgage of £283,275.00 at 6.4% annually for 20 years. Calculate the monthly payment and total interest paid.

Answer:

Monthly payment: £2,095.38, total interest: £219,616.20

Working:

Monthly rate = 6.4% ÷ 12 = 0.5333% = 0.005333

Number of payments = 20 × 12 = 240

$(1 + r)^n = (1 + 0.005333)^{240} = 3.5844$

$M = P \times [r(1+r)^n] \div [(1+r)^n - 1]$

$M = £283,275.00 \times [0.005333 \times 3.5844] \div [3.5844 - 1]$

Monthly payment = £2,095.38

Total paid = £2,095.38 × 240 = £502,891.20

Total interest = £502,891.20 - £283,275.00 = £219,616.20

E2

An investment of £41,382.00 grew to £94,384.31 over 6 years. Calculate the total return percentage and annual return rate.

Answer:

Total return: 128.08%, annual return: 14.73%

Working:

Total return = £94,384.31 - £41,382.00 = £53,002.31

Total return % = (£53,002.31 ÷ £41,382.00) × 100 = 128.08%

Annual return = $[(£94,384.31 \div £41,382.00)^{1 \div 6} - 1] \times 100$

Annual return = 14.73%

Teacher answers & working

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E3

£5,801.00 is invested at 4.92% compound interest for 10 years. Find the final amount and interest earned.

Answer:

Final amount: £9,377.47, interest: £3,576.47

Working:

Use $A = P(1 + r)^t$.

$P = £5{,}801.00$, $r = 0.0492$, $t = 10$

Growth multiplier = 1 + 0.0492 = 1.0492

$A = £5{,}801.00 \times (1.0492)^{10}$

$A = £5{,}801.00 \times 1.6165 = £9{,}377.47$

Interest = £9,377.47 - £5,801.00 = £3,576.47